

Continuous Integration with Salesforce DX:

Practices and Principles for All

David Reed

PhillyForce, April 30, 2018 | #PhillyForce18

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David Reed

- Senior Salesforce Developer, Radian
- Certified Platform Developer II
- Open source code at github.com/davidmreed
- Interested in Python, C, Ancient Greek, roller derby.



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Platform Developer II

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Platform Developer I

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Administrator

Agenda

1. What is Continuous Integration?
 - a. A Conceptual Overview
 - b. Major Benefits
2. Setting Up CI with Salesforce DX and CircleCI
 - a. Core Setup for Any Project
 - b. Building Integration and Test Flow
3. Putting It All Together
 - a. Reviewing a Complete Workflow
 - b. Getting and Using Results
4. Taking the Next Steps
 - a. Adding Concepts and Tools
 - b. Links and Resources

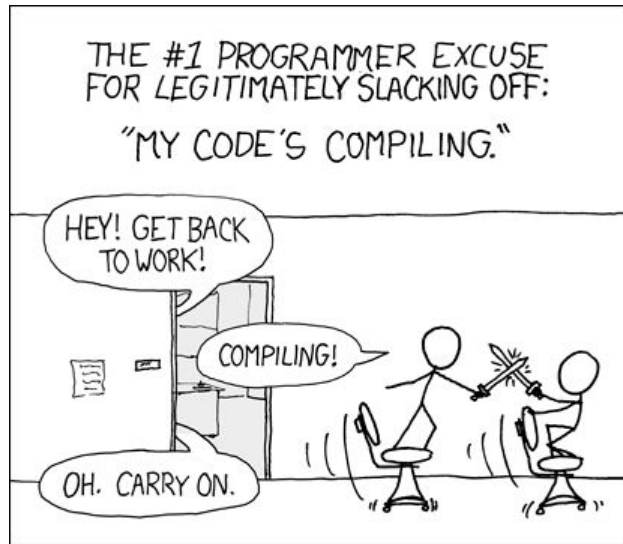
What is Continuous Integration?

A Conceptual Overview

This is the “Principles” part.

Introduction: What Is Continuous Integration?

- Build and test your software constantly and automatically.
- Use scripting to reduce or eliminate human intervention.
- Add tools to increase insight and value



Your code is never **not** compiling.

CI is for...

- Developers.
- Admins who work in an org that has code.
- Implementors and solution architects in code or code/declarative orgs.
- Release and application lifecycle managers.
- Leadership overseeing development activities.

The Foundations

- Salesforce DX scratch orgs:
work and testing location
- Source code: source of truth

Commits on Apr 14, 2018

Fix test case
 davidmreed committed a day ago ✓
Add admin work
 davidmreed committed a day ago ✗
Update unit tests
 davidmreed committed a day ago ✓

Truth



History



```
example-project
├── .circleci
│   └── config.yml
├── .sfdx
├── assets
├── config
├── force-app
│   └── main
│       ├── default
│       │   ├── aura
│       │   └── classes
│       │       ├── AccountTriggerHandler.cls
│       │       ├── AccountTriggerHandler.cls-meta.xml
│       │       └── AccountTriggerHandlerTest.cls
│       │           ├── AccountTriggerHandlerTest.cls-meta.xml
│       │           └── objects
│       └── triggers
│           ├── AccountTrigger.trigger
│           └── AccountTrigger.trigger-meta.xml
├── .forceignore
├── .gitignore
├── README.md
└── sfdx-project.json
```

What Does CI Flow Look Like?

Write Code ➡ Push ➡ Test ➡ Feedback

Create Scratch Org
Push Source
Run Apex Tests
Delete Scratch Org

```
$ git push
```



Test Summary

Queue (00:00)

Your build ran **27** tests in unknown with **0 failures**

Slowest test: MyProfilePageControllerTest testSave (took 0.22 seconds).

Show containers: All (1) Successful (1) Failed (0)

0
(00:59)

Major Benefits

"If I've made an error in my updates, correcting and testing multiple versions of an app takes minutes, not hours or days."

— Zayne Turner, "Getting Started with Salesforce DX"

Visibility

Continuous integration gives you visibility across your codebase, every commit.

- Does it compile?
- Tests passing?
- Regressions?
- With integrated tools:
 - Overall test coverage?
 - Coding best practices?

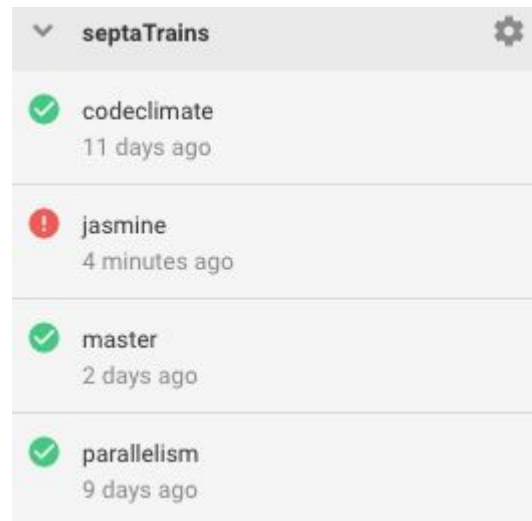
All checks have passed

4 successful checks

✓		ci/circleci: build — Your tests passed on CircleCI!	Details
✓		ci/circleci: static-analysis — Your tests passed on ...	Details
✓		codecov/patch — Coverage not affected when com...	Details
✓		codecov/project — 100% remains the same compa...	Details

Risk Reduction

- Never unknowingly merge broken code.
- Find more regressions faster - no waiting on manual test cycles.
- Surface areas of risk.
- Iterate and branch without chaos.



A feature branch is failing, but master is green.

Virtuous Feedback Cycle

- Utility of CI goes up with good development practices.
- Value of good practices increases and becomes visible with CI.



Setting Up CI with Salesforce DX and CircleCI

This is the “Practices” part.

How Are We Doing This?

- Stepping through setup process.
- Don't sweat the details. Initial setup is largely boilerplate.
- See how concepts are applied and systems connected.

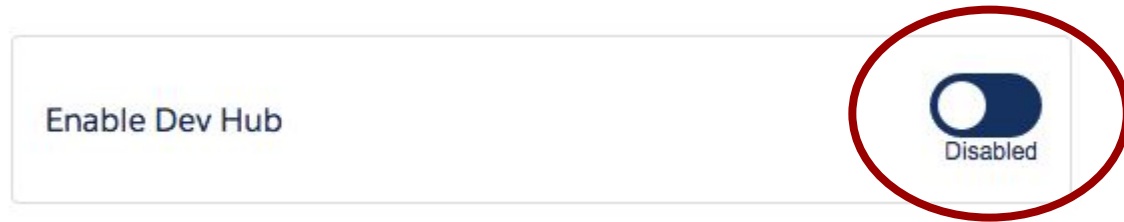
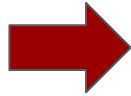
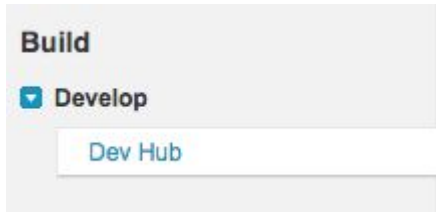


Warning: technical content ahead!

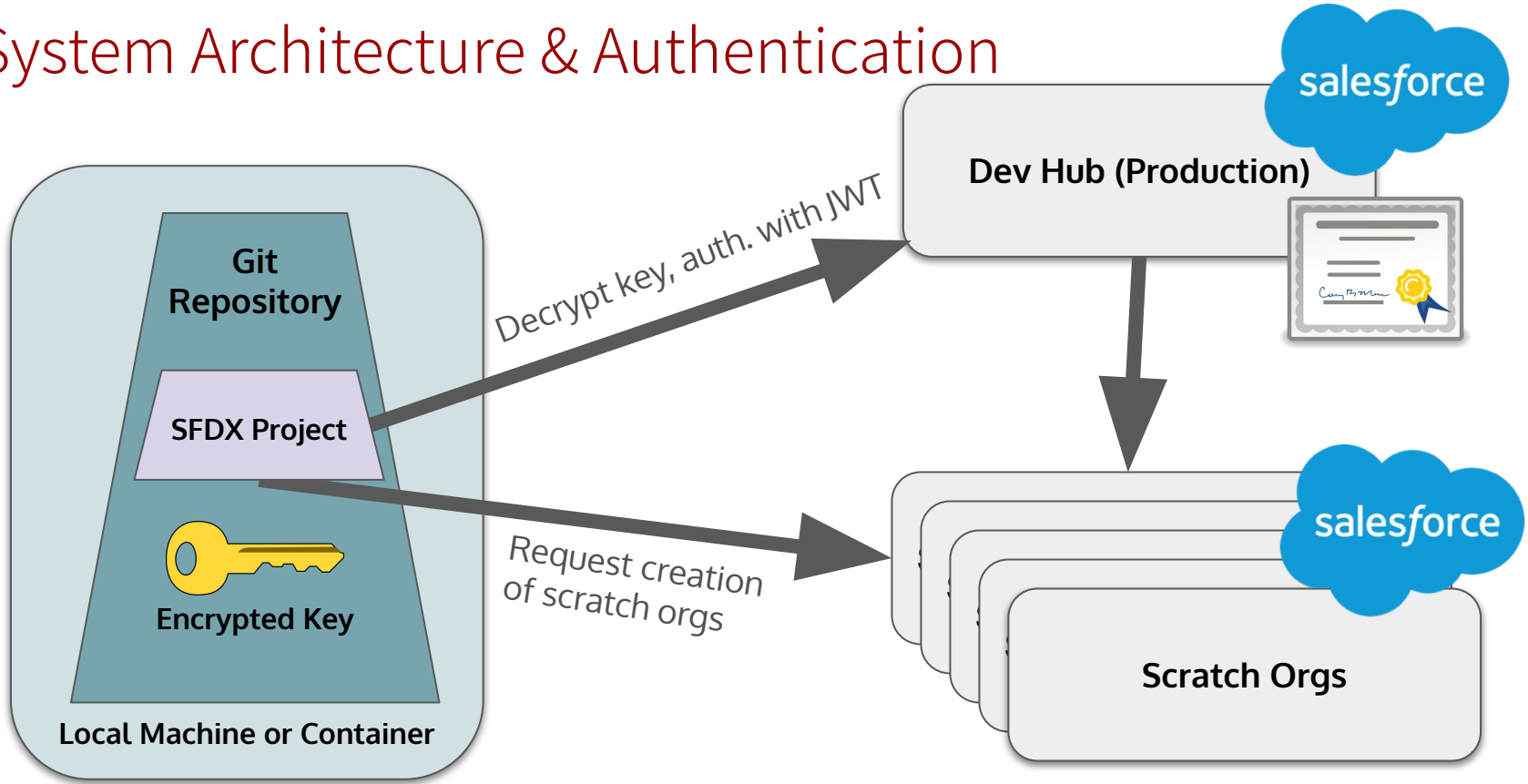
Core Setup for Any Project

Activate Dev Hub to Manage Scratch Orgs

- Production org or Dev Hub trial.
- **No effect** on other production functionality.



System Architecture & Authentication



Establish Project Structure

```
$ sfdx force:project:create  
--projectname proj$ sfdx  
force:auth:web:login -a DevHub -d  
$ cd proj  
$ git init .  
$ git remote add origin  
<GITHUB_URL>  
$ git add force-app && git commit  
-m "Initial commit"
```

- Create SFDX project.
- Authorize Dev Hub using OAuth web flow.
- Move to project directory.
- Initialize Git repository.
- Add GitHub project URL.
- Commit the initial project structure.

Generate Keys and Certificate

```
$ openssl genrsa -des3 -passout  
pass:x -out server.pass.key 2048  
$ openssl rsa -passin pass:x -in  
server.pass.key -out server.key  
$ rm server.pass.key  
$ openssl req -new -key server.key  
-out server.csr  
$ openssl x509 -req -sha256 -days  
365 -in server.csr -signkey  
server.key -out server.crt  
$ rm server.csr
```

- Create key pair and certificate
- Generic commands for any project
- Keep your certificate file safe - do not commit it to Git!

Encrypt and Commit Key

```
$ mkdir assets
$ mv server.key assets
$ openssl aes-256-cbc -k $KEY -in
assets/server.key -out
assets/server.key.enc -e
$ rm assets/server.key
$ git add assets
$ git commit -m "Add encrypted key"
```

- Replace \$KEY with your secure password
- Use a password manager!
- We're committing only the encrypted keyfile to Git - plaintext file is deleted.
- Your password will go in CircleCI environment variables.

Products



Git Repo / SFDX Project

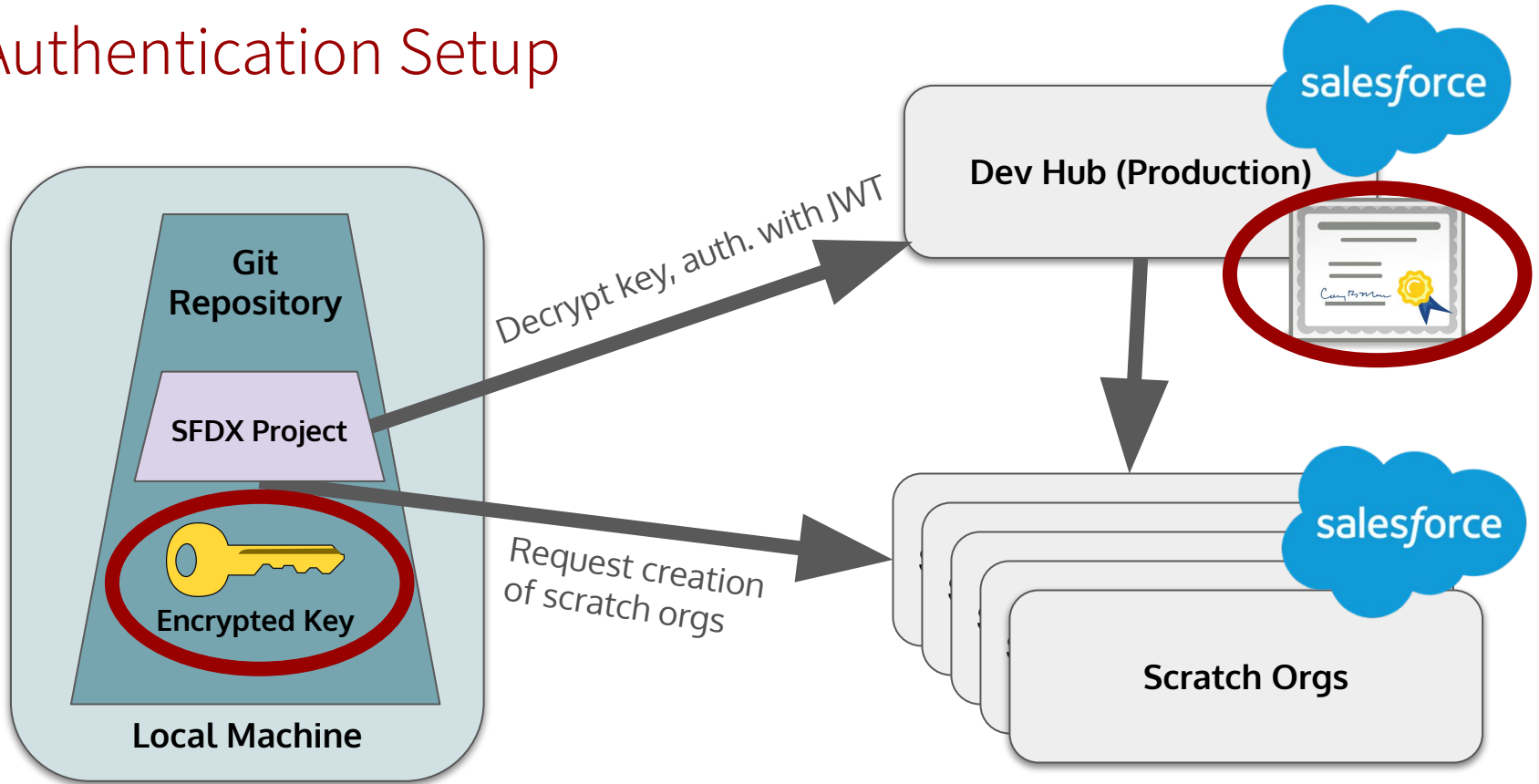


Encrypted Key File
`server.key.enc`



Server Certificate
`server.crt`

Authentication Setup



Build Connected App in Dev Hub Org

Connected App Name
SFDX CLI

[« Back to List: Custom Apps](#)



[Edit](#) [Delete](#) [Manage](#)

Version	1.0
API Name	SFDX_CLI
Created Date	1/29/2018 12:28 PM
	By: David Reed
Contact Email	david@ktema.org
Contact Phone	
Last Modified Date	2/2/2018 5:55 AM
	By: David Reed
Description	
Info URL	

- Save Consumer Key securely.
- Note OAuth scopes and Callback URL.
- Certificate provided on creation.

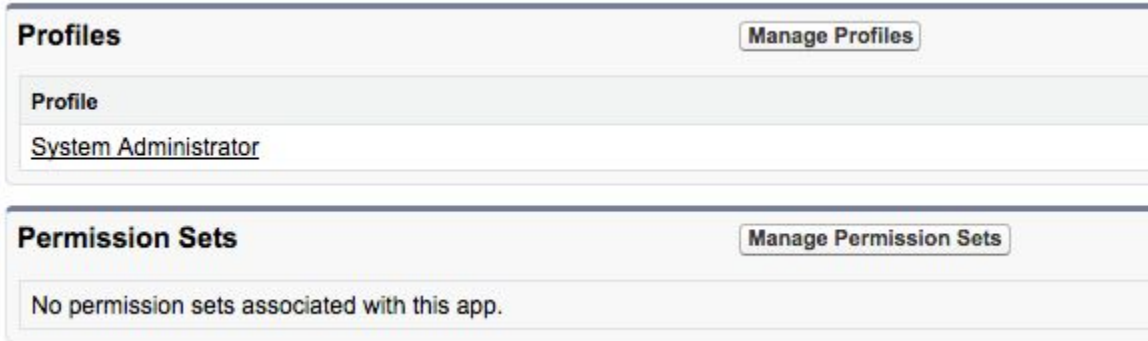
▼ API (Enable OAuth Settings)

Consumer Key	(value hidden)	Consumer Secret	Click to reveal
Selected OAuth Scopes	Access and manage your data (api) Provide access to your data via the Web (web) Perform requests on your behalf at any time (refresh_token, offline_access)	Callback URL	http://localhost:1717/OauthRedirect
Digital Certificate	O=Ktema Systems, L=Philadelphia, ST=PA, C=US 11 Dec 2018 21:35:36 GMT	Enable for Device Flow	☐
Require Secret for Web Server Flow	☒	Token Valid for	0 Hour(s)
Include Custom Attributes	☐	Include Custom Permissions	☐
		Enable Single Logout	Single Logout disabled

Authorize Users to Connected App



- Preapproved users.
- Add profile or permission set for Continuous Integration user.

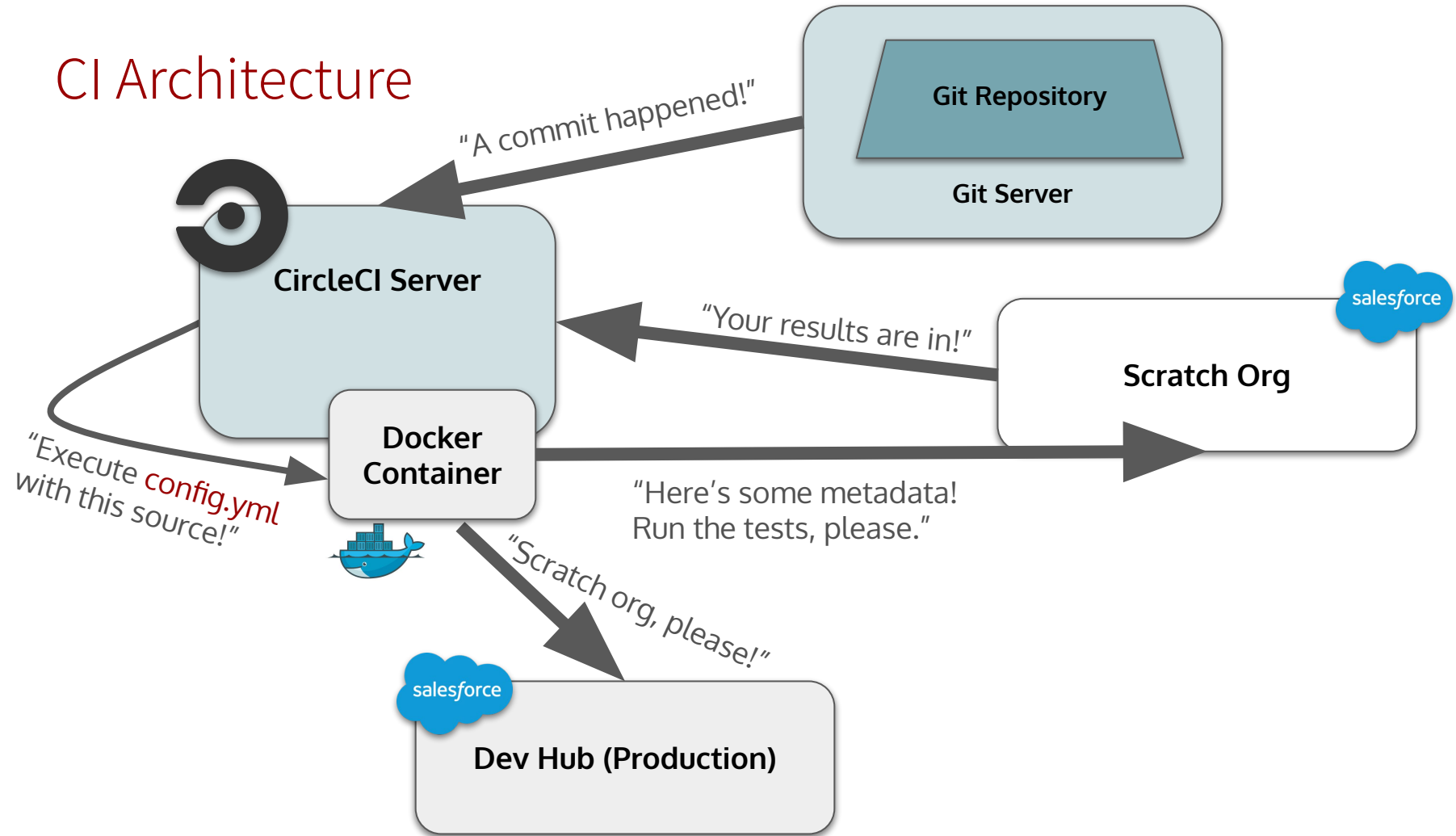


Did You Get All of That?

- It's okay if you didn't.
- This is boilerplate setup.
- Each command is not relevant to the practices and principles of continuous integration.
- What's critical is *how* and *when* systems communicate.

Building Integration and Test Flow

CI Architecture



Build config.yml

- `config.yml` is the driver of the CI flow. (Similar files for Travis, Jenkins, etc.)
- Complete templates available on GitHub (github.com/davidmreed).
- “Expand the umbrella” - learn tools outside Salesforce.



%YAML 1.2

YAML: YAML Ain't Markup Language



Prepare CI Image Environment

- run:

name: Install Salesforce DX

command: |

```
if [ ! -d node_modules/sfdx-cli ]; then
  export SFDX_AUTOUPDATE_DISABLE=true
  export SFDX_USE_GENERIC_UNIX_KEYCHAIN=true
  export SFDX_DOMAIN_RETRY=300
  npm install sfdx-cli
  node_modules/sfdx-cli/bin/run --version
  node_modules/sfdx-cli/bin/run plugins --core
fi
```

- run:

name: Decrypt Server Key

command: |

```
openssl aes-256-cbc -k $KEY -in assets/server.key.enc -out assets/server.key -d
```

- **run** steps execute in container.
- Using a **node.js** base image.
- Install Salesforce DX and decrypt our key to connect to Dev Hub.

Create New Scratch Org

```
- run:
  name: Create Scratch Org
  command: |
    node_modules/sfdx-cli/bin/run force:auth:jwt:grant --cli
    entid $CONSUMERKEY --jwtkeyfile assets/server.key --username $USERNAME
    --setdefaultdevhubusername -a DevHub
    node_modules/sfdx-cli/bin/run force:org:create -v DevHub
    -s -f config/project-scratch-def.json -a scratch
```

- Authenticate to Dev Hub with JWT (headless) flow - using private key.
- Request a new scratch org via DX.
- DX holds reference to scratch org.

Push Source and Run Tests

```
- run:
  name: Push Source
  command: |
    node_modules/sfdx-cli/bin/run force:source:push -u scratch
- run:
  name: Run Apex Tests
  command: |
    mkdir ~/apex_tests
    node_modules/sfdx-cli/bin/run force:apex:test:run -u scratch -c -r human -d ~/apex_tests
```

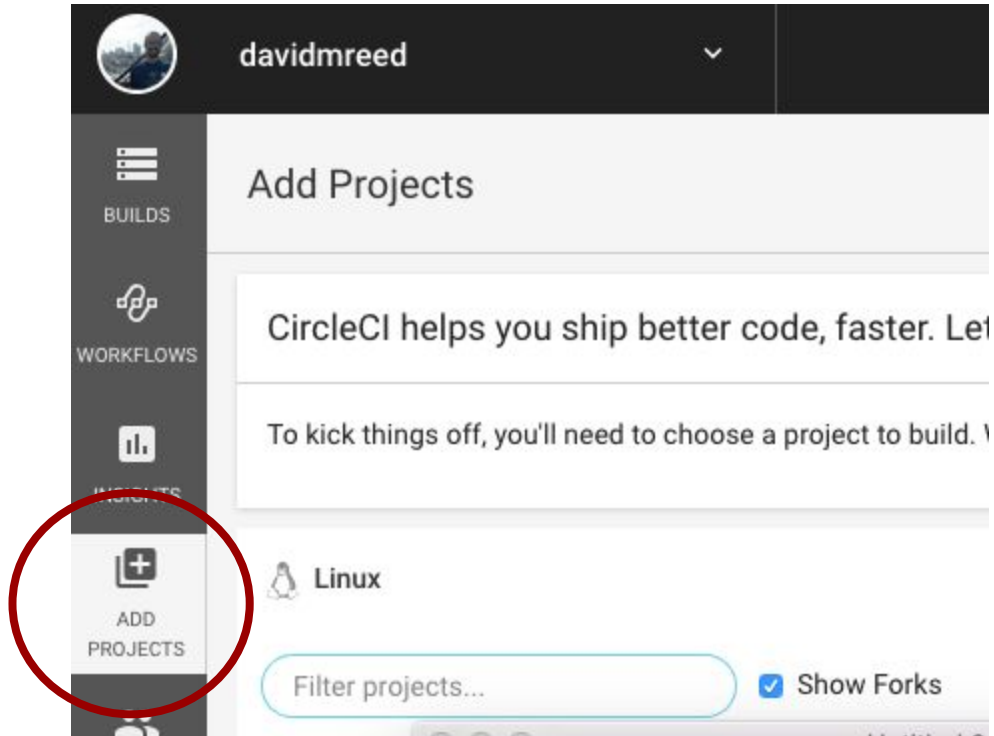
- Push source to scratch org.
 - Will fail on syntax/reference errors.
 - *All* code and metadata pushed.
 - *Zero* effect on Dev Hub (except limits).
- Run tests and wait for completion.

Clean Up

```
- run:  
  name: Clean Up  
  when: always  
  command: |  
    node_modules/sfdx-cli/bin/run force:org:delete -u scratch -p
```

- We immediately delete scratch org; Docker environment is disposed.
- Only artifact is test results stored in CircleCI.
- Zero changes to prod/sandbox.
- Keep scratch org limits in mind.

Add Project to CircleCI



Configure CircleCI Environment Variables

Store keys and server settings securely in CircleCI, not in Git.

Environment Variables for davidmreed/septaTrains

Import Variable(s)

Add Variable

Add environment variables to the build. You can add sensitive data (e.g. API keys) here, rather than placing them in the repository. The values can be any bash expression and can reference other variables, such as setting `M2_MAVEN` to `${HOME}/.m2`.

Name	Value	Remove
CONSUMERKEY	xxxx	×
KEY	xxxx	×
USERNAME	xxxx	×

Putting It All Together

This is the “Practices and Principles” part.

Reviewing a Complete Workflow

Write Code

```
@isTest
public static void testThe
{
    System.assert(false);
}
```

Regression!



```
@isTest
public static void testThe
{
    System.assert(true);
}
```

Apply Fix

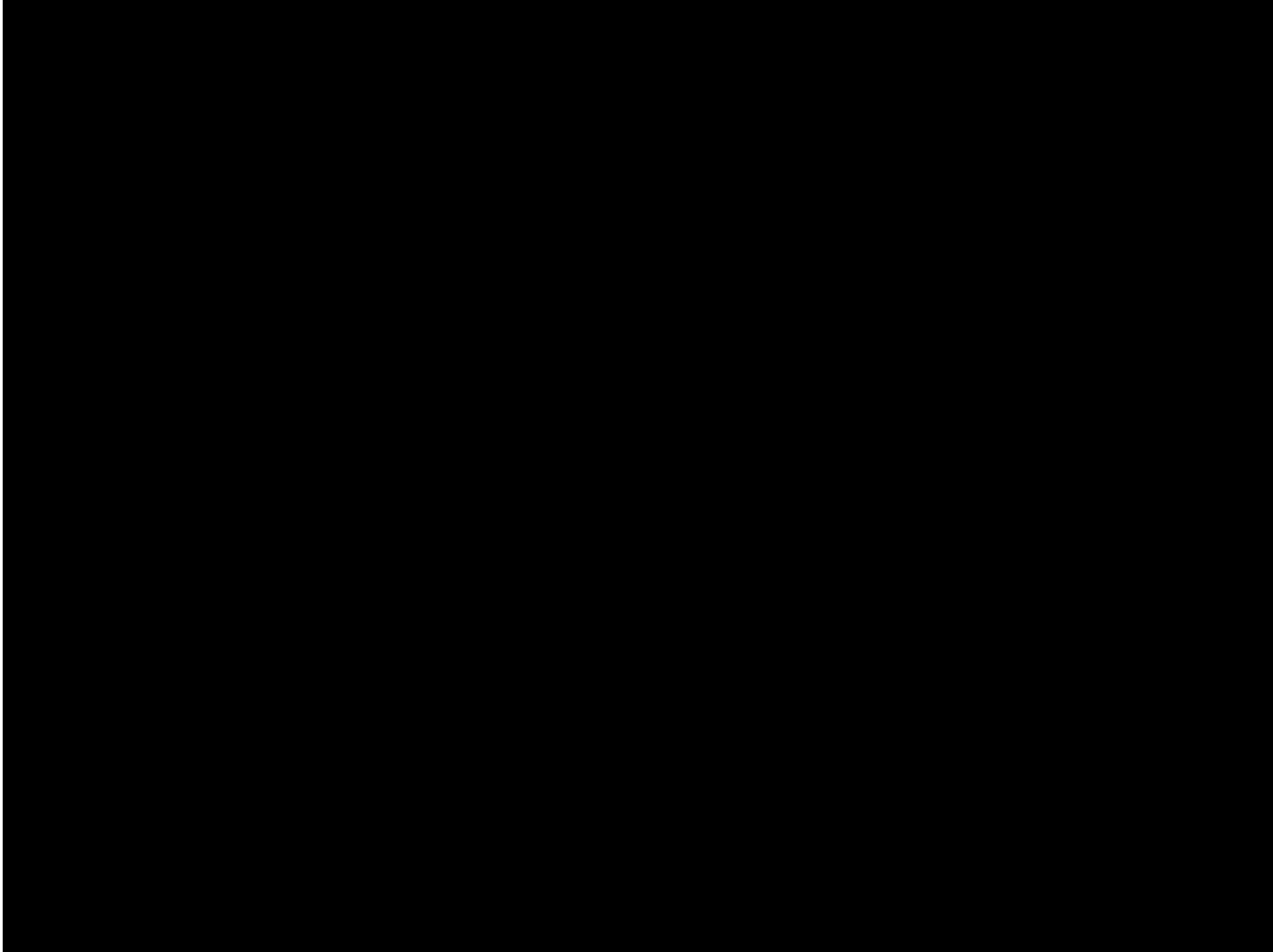


Admin Changes

Validation Rule Edit

Rule Name	New_Rule
Active	☒

Ready for Prod



Getting and Using Results

Build and Test Reporting

FIXED

Finished:

2 months ago (02:06)

Previous:

15

Parallelism:

1x out of 4x

Queued:

00:00 waiting + 00:02 in queue

Resolved:

2CPU

COMMITTS (1)

David Reed

GitHub

fddce46

Delete unencrypted server key as early as possible

Test Summary

Queue (00:02)

Artifacts

Your build ran **8** tests in unknown with **0 failures**

Slowest test: DMRImportNotesPageControllerTEST testImportOperation (took 24.34 seconds).

Show containers:

All (1)

Successful (1)

Failed (0)

0

(02:06)

- septaTrains**
- codeclimate

11 days ago
- jasmine

4 minutes ago
- master

2 days ago
- parallelism

9 days ago

GitHub Status Updates

davidmreed / **DMRNoteAttachmentImporter**

<> Code ! Issues 0 Pull requests 0 Projects

Branch: master ▼

Commits on Jan 31, 2018

Delete unencrypted server key as early as possible
davidmreed committed on Jan 31 ❌

Commits on Dec 21, 2017

Add coverage badge
davidmreed committed on Dec 21, 2017 ✅

Every commit is tagged with a build status.

All checks have passed

4 successful checks

- ✅  **ci/circleci: build** — Your tests passed on CircleCI!
- ✅  **ci/circleci: static-analysis** — Your tests passed on ...
- ✅  **codecov/patch** — Coverage not affected when com...
- ✅  **codecov/project** — 100% remains the same compa...

Detailed results of all jobs and checks on each commit.

Live Build Notifications

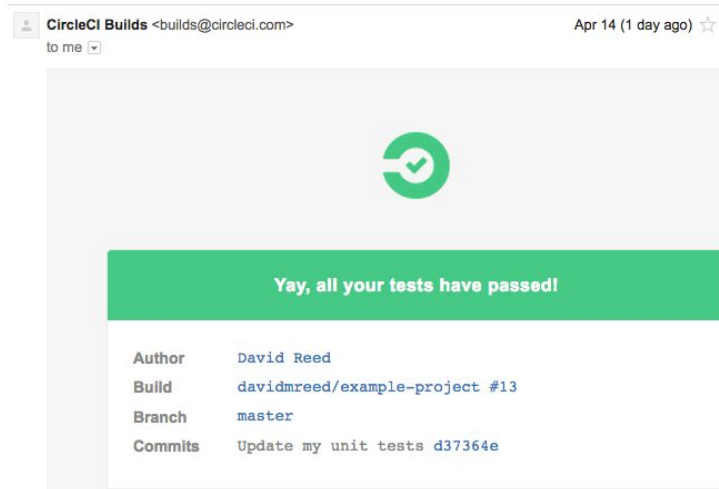
 **circleci** 4:58 PM
Hi, welcome to Circle! Slack notifications are now enabled for [my-team/awesome-project](#)

 **circleci** 5:06 PM
Failed in build [#127](#) of [my-team/awesome-project](#) (master)
- jchan: Missing file ([5ec4a23](#))

 **circleci** 5:17 PM
Fixed in build [#128](#) of [my-team/awesome-project](#) (master)
- jchan: Fixed file pointer ([b443af5](#))

Slack (image via CircleCI)

Build emails



Web notifications

Taking the Next Steps

Adding Concepts and Tools



Key Concepts

- Once you're scripting with CI, you have enormous power to add tools.



- `run` steps are just shell scripts.

- Docker virtualization puts almost any software in reach - just

`apt-get` or `npm install`.



mercurial

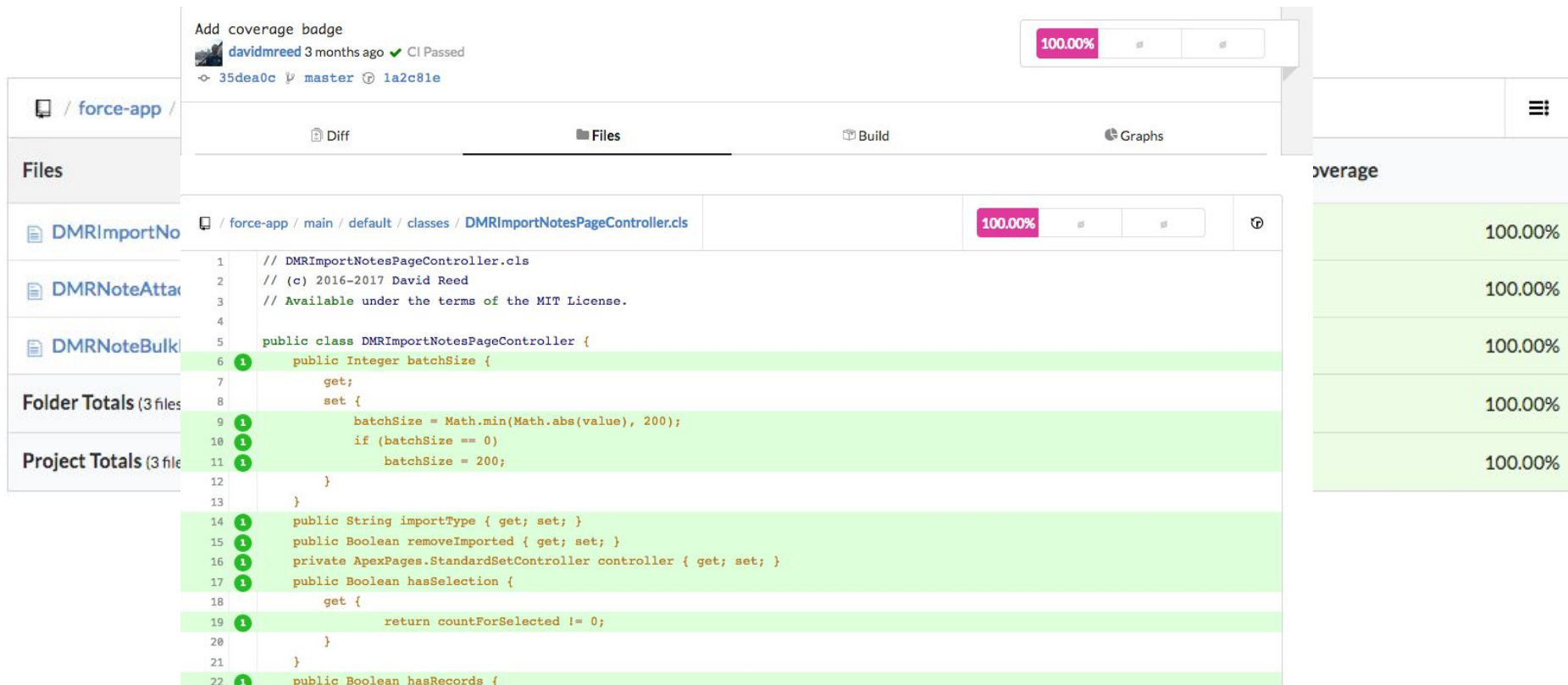
- Moving fluidly between tools to get the job done.
- Taking advantage of prior art, then innovating.



Travis CI



Visualization of Code Coverage with Codecov.io



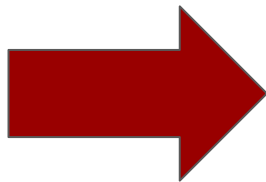
Testing Multiple Org Types and Shapes

```
{ } project-scratch-def.json ●  
1 {  
2   "orgName": "David Reed",  
3   "edition": "Enterprise",  
4   "features": ["Sites"],  
5   "orgPreferences" : {  
6     "enabled": ["S1DesktopEnabled"]  
7   }  
8 }  
9
```

Scratch definition JSON file(s)

```
workflows:  
  version: 2  
  test_and_static:  
    jobs:  
      - build-enterprise  
      - build-developer  
      - static-analysis
```

Workflow definition

A screenshot of a CircleCI workflow run. At the top, a green banner indicates the workflow 'SUCCEEDED'. Below this, the branch 'master / test_and_static' is shown, along with a link to 'Move to SFDX 6.8.2'. A 'Rerun' button is visible. The workflow consists of three jobs, all of which are marked as successful with green checkmarks and their respective durations: 'build-enterp...' (01:32), 'static-analy...' (00:06), and 'build-develo...' (01:44).

master / test_and_static
Move to SFDX 6.8.2
Rerun

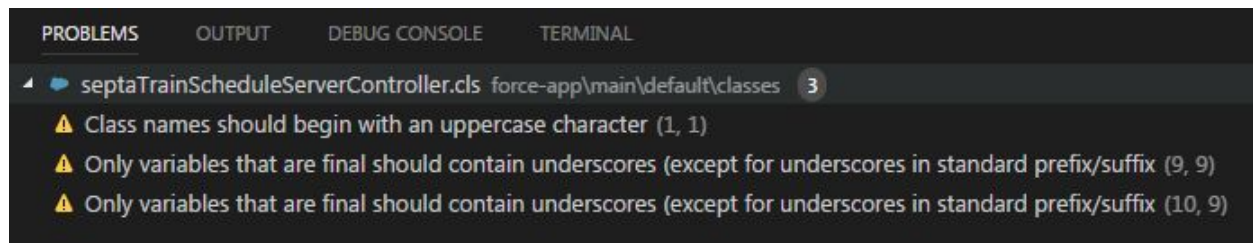
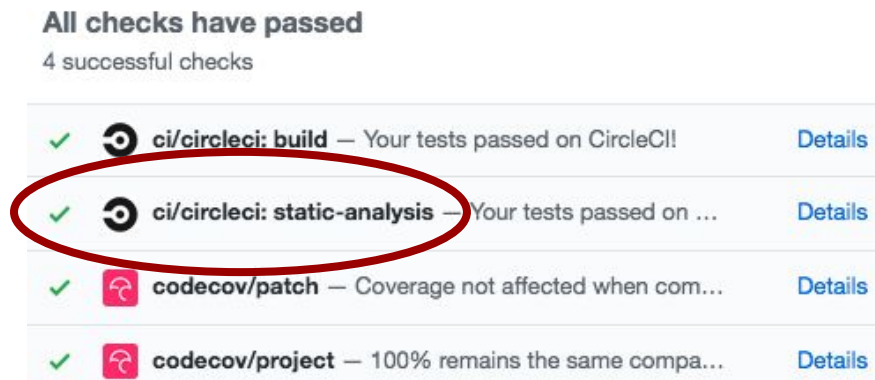
3 jobs in this workflow

- ✓ build-enterp... ⌚ 01:32
- ✓ static-analy... ⌚ 00:06
- ✓ build-develo... ⌚ 01:44

Results per org type in CircleCI and GitHub

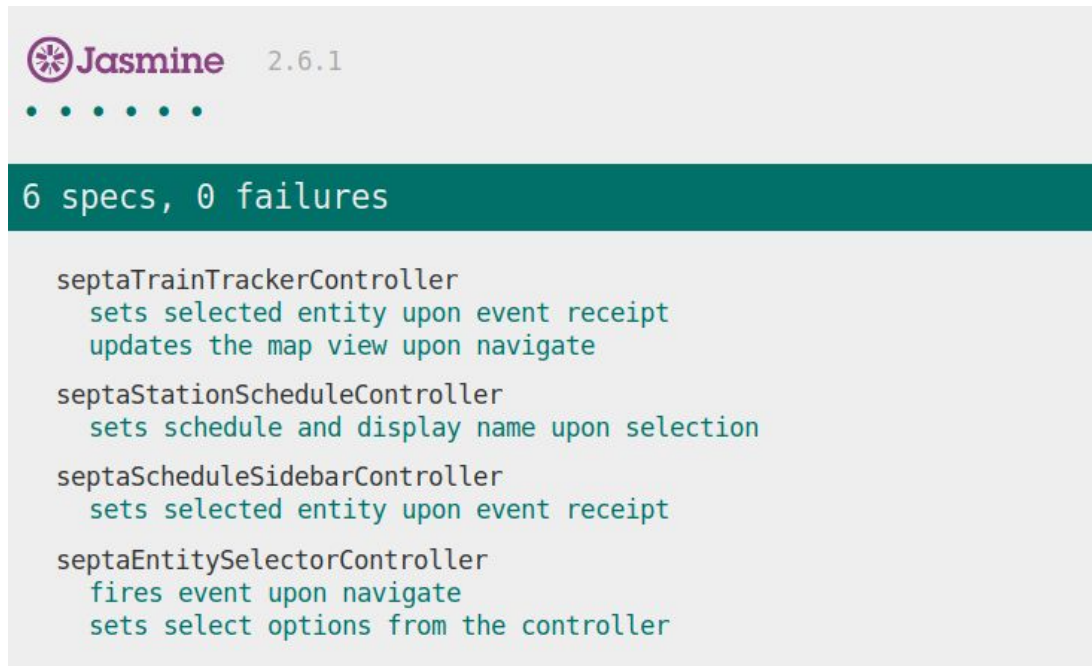
Static Analysis with PMD

- PMD static analysis can run *in parallel* with scratch org deploy and test.
- Block the build on SOQL/DML in loops, more.
- PMD can also run in your IDE.



Lightning Testing Service

- Run JavaScript unit tests, using Mocha or Jasmine, in Salesforce DX and CI.



The image shows a screenshot of the Jasmine test runner interface. At the top left is the Jasmine logo (a purple flower-like icon) followed by the text "Jasmine" and the version number "2.6.1". Below this is a row of six small blue dots. A dark green horizontal bar contains the text "6 specs, 0 failures" in white. Below the bar, the test results are listed in a light gray area. The results are organized into four groups, each starting with a controller name followed by its specifications:

```
septaTrainTrackerController
  sets selected entity upon event receipt
  updates the map view upon navigate

septaStationScheduleController
  sets schedule and display name upon selection

septaScheduleSidebarController
  sets selected entity upon event receipt

septaEntitySelectorController
  fires event upon navigate
  sets select options from the controller
```



Testing Software that Integrates with Salesforce

- Salesforce DX isn't limited to Apex code running on the Force.com platform.
- Code that integrates to Salesforce can be tested against scratch orgs too.

```
- run:
  name: Integration Test - Username and Password
  command: |
    . venv/bin/activate
    node_modules/sfdx-cli/bin/run force:user:password:generate > /dev/null
    export PASSWORD=$(node_modules/sfdx-cli/bin/run force:user:display --json | pyth
    export SF_USERNAME=$(node_modules/sfdx-cli/bin/run force:user:display --json | p
    python example-simple-salesforce.py -p "$PASSWORD" -u "$SF_USERNAME" -t "" -s
```

Integration testing a Python/simple_salesforce application

Links and Resources

- Salesforce DX: developer.salesforce.com/platform/dx
- Dev Hub Trial Org: developer.salesforce.com/promotions/orgs/dx-signup
- Getting Started with Salesforce DX (5-part series):
developer.salesforce.com/blogs/2018/02/getting-started-salesforce-dx-part-1-5.html
- Continuous Integration with Salesforce DX Trailhead:
trailhead.salesforce.com/en/modules/sfdx_travis_ci
- Templates on GitHub: github.com/davidmreed/circleci-sfdx-examples
- Travis Example Repo: github.com/forcedotcom/sfdx-travisci

Thank you!

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